# Assylbek S. Gizatov

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## Education

**Astana IT University**, Astana, Kazakhstan — September 2022 – June 2025
Bachelor of Science in Information Technologies
Specialization: Mathematical and Computational Science

**Nazarbayev Intellectual School**, Atyrau, Kazakhstan — September 2016 – June 2022
High School Diploma, A-Level

## Academic Positions

**BrainCode Academy** — October 2026 – April 2027
**Lecturer in Machine Learning and Artificial Intelligence**
Tech Orda Program, Astana Hub
Government-supported IT education initiative implemented by Astana Hub with the support of the Ministry of Artificial Intelligence and Digital Development of the Republic of Kazakhstan.

**Kh. Dosmukhamedov Atyrau University**, Atyrau, Kazakhstan — July 2025 – Present
**Senior Researcher**
Research Institute of Physics, Mathematics and Applied Technologies
Director: Zhusupkalieva Galiya, Ph.D.

**Kh. Dosmukhamedov Atyrau University**, Atyrau, Kazakhstan — January 2026 – June 2026
**Lecturer**
Department of Software Engineering, Faculty of Physics, Mathematics and Information Technology
Dean: Baktygul Assanova, Ph.D.

**Astana IT University**, Astana, Kazakhstan — September 2024 – June 2025
**Research Assistant, AI & Smart City**
Supervisor: Akmaral Kuatbayeva, Ph.D.

**Chongqing City Vocational College**, Chongqing, China — December 2024
**Research Program, UAV/UAS**
Supervisor: Zarina Kutpanova, Ph.D.

## Industry Experience

**North Caspian Operating Company N.V.** — August 2025 – Present
**AI Technology & Enablement Consultant**

- Analyzed 33+ AI use cases consolidated from 9 business directorates, including POD, TD, and NDD, to support implementation planning, licensing assessments, and capacity planning.
- Established the AI tokenomics approach by assessing architecture complexity, adoption scenarios, orchestration, reasoning models (GPT, Claude Sonnet, Opus, Fable), and projected consumption patterns to support 2026–2027 investment and capacity planning.
- Developed and formalized AI regulatory and governance documentation, establishing the foundation for NCOC's enterprise AI governance and regulatory framework.
- Led pilot-readiness and assessment activities for Microsoft 365 Copilot, Copilot Teams, Copilot Notebook, Copilot agents, and Copilot Co-work.
- Led the internal launch and presentation of Microsoft Copilot Studio, introducing its capabilities and adoption opportunities across the company.
- Served as the focal point for Copilot Studio-based AI Champions initiatives, supporting 10+ use cases and advising teams across Strategy, Government Relations, Legal, and other departments, including the Senior Leadership Team.
- Pioneered and deployed AI-Sana, NCOC's SLT-endorsed flagship AI agent, serving 5,000+ users across NCOC and global partner organizations including Shell, ExxonMobil, and TotalEnergies.
- Developed AI Transcriber for Zoom, a local speech-recognition application powered by the GigaAM 220M CTC model, enabling colleagues to generate transcripts.
- Developed a multi-tool PMIS AI Agent integrating 15+ Power Automate flows and Dataverse data sources to provide relationship-aware project insights across tasks, risks, issues, costs, and related records.
- Automated project reporting and status consolidation, reducing manual effort by 200+ hours per month.
- Collaborated with service providers and technology vendors on the evaluation, implementation, and alignment of new AI platforms and tools, supporting enterprise adoption and integration.
- Built internal AI capability by coaching cross-functional teams on AI agent architecture, workflow automation, orchestration, and responsible AI practices using the Microsoft ecosystem.
- Received a Letter of Recognition from the Technical Director for contributions to enterprise AI and IT operations.

**BinY.Co** April 2025 – September 2025
**Middle Data Scientist**

- Freight-tech startup raised approximately $1 million at a $2.5 million valuation, backed by a McKinsey & Company Managing Partner.
- Developed foundational mathematical, pricing, and AI models.
- Engineered autonomous data-acquisition pipelines using Crawl4AI with deep-link discovery and multi-level web crawling, transforming unstructured web data into production-ready datasets.
- Developed AI-powered document intelligence and analytics systems using Docling, RAG architectures, Pydantic structured outputs, and pricing-optimization models achieving 95%+ predictive accuracy.

## Publications

### Peer-reviewed Conference Articles

C1. I. Orynbassar, **A. Gizatov**. *Cross-Dataset Generalization in ECG Arrhythmia Classification: Quantifying Two Sources of Over-Optimism*. 2026 IEEE 3rd International Conference on Digital Generation (DG), 2026.

C2. I. Orynbassar, **A. Gizatov**, N. Kituyev, B. Kenzheakhmetov. *Aqyl Jol: An AI-Based System for Urban Traffic Management and Municipal Fleet Coordination*. 2026 IEEE 6th International Conference on Smart Information Systems and Technologies (SIST), 2026. doi:10.1109/SIST61674.2026.11595985 .

C3. **A. Gizatov**, N. Kituyev, I. Orynbassar, G. Amangeldy, A. Zhannurova. *Application of Causal Machine Learning to Compute the Effects of Video Game Intensity on Students' Academic Performance and Psychosocial Well-Being*. DiGRA Digital Library, 2026. doi:10.26503/dl.v2026i2.2780 .

C4. **A. Gizatov**, N. Zhaksybayev. *Predictive Models for the Labour Market of Kazakhstan*. International Scientific Conference "Digital Qazaqstan: Reality and Prospects in the Era of Artificial Intelligence", pp. 320–325, 2025.

C5. **A. Gizatov**, A. Kamit, A. Kandyshev, E. Kapsamatov. *Causal Machine Learning for Digital Education and Service Sector: Predicting Engagement and Optimizing Interventions*. International Scientific Conference "Digital Qazaqstan: Reality and Prospects in the Era of Artificial Intelligence", pp. 314–320, 2025.

C6. **A. Gizatov**, I. Orynbassar, G. Amangeldi. *Intelligent Credit Scoring System Based on Machine Learning for Creditworthiness Assessment and Personalization of Financial Decisions*. International Scientific Conference "Digital Qazaqstan: Reality and Prospects in the Era of Artificial Intelligence" (Raimov Forum), pp. 307–314, 2025.

C7. A. Kuatbayeva, M. Sergaziyev, D. Yedilkhan, **A. Gizatov**, D. Issenov, A. Namet, O. Bekbolatov. *Design ML Models for Bus Time Arrival Prediction in Astana City*. International Conference on "Artificial Intelligence and Inverse Problems in Science, Technology and Industry", pp. 9–12, 2025.

C8. **A. Gizatov**, A. Kuatbayeva, A. Jangeldin, N. Zhaksybayev. *Predictive Models for Analyzing and Forecasting Labor Market Trends in Kazakhstan: Addressing Market Saturation and Ensuring Economic Stability*. International Conference on "Artificial Intelligence and Inverse Problems in Science, Technology and Industry", pp. 217–220, 2025.

### Peer-reviewed Journal Articles

J1. **A. Gizatov**, N. Zhaksybayev, A. Jangeldin. *Multi-Modal Ensemble Learning Architecture for Labor Market's Digital Transformation*. Energy ELEMENTS Journal, no. 1, pp. 91–118, 2026. doi:10.68458/ee.2026.1.6 .

J2. A. Kuatbayeva, M. Sergaziyev, D. Yedilkhan, D. Issenov, **A. Gizatov**. *Multi-Output Bus Travel Time Prediction Using Convolutional LSTM Neural Networks*. Scientific Journal of Astana IT University, vol. 22, pp. 189–205, 2025. doi:10.37943/22JALW4601 .

J3. **A. Gizatov**, I. Orynbassar, G. Amangeldi, B. Assanova. *Intelligent Credit Scoring System Based on Machine Learning for Creditworthiness Assessment and Personalization of Financial Decisions*. Banks of Kazakhstan Journal, nos. 11–12, pp. 72–73, 2025.

J4. **A. Gizatov**, A. Kuatbayeva, A. Kandyshev. *Development of Models for Analyzing and Forecasting the Profitability of Key Professions to Prevent Oversaturation of the Labor Market in Kazakhstan*. Ecology and Sustainable Development, pp. 41–48, 2025.

**Entrepreneurial Projects**

**Aqyl Jol, accelerated by Meta**: *AI system for urban traffic management and municipal fleet coordination*.
**Co-founder & CEO**

- Founded and led an AI-powered Smart City initiative focused on traffic optimization and municipal route planning.
- Managed end-to-end project delivery, including requirements gathering, stakeholder engagement, risk management, and executive reporting.
- Estimated economic benefits of more than $1 million through urban traffic and route optimization.
- Won 1st place in the 2025 Meta Accelerator Program, ranking #1 among 1,000+ participants.
- Won 2nd place at the Startup Battle during the 2026 International Forum "Digital Atyrau."
- Secured $7,000 in AWS cloud credits and infrastructure resources to support AI model development, testing, and deployment.
- Published the project's scientific contribution and AI methodology in an IEEE research paper.
- Presented the solution to representatives of Meta, AWS, and government organizations at MLA Demo Day during AI & Digital Bridge 2025.
- Selected by Kazakhstan's Ministry of Science and Higher Education to showcase the project at Startup Alley during AI & Digital Bridge 2026, with exhibition and participation costs fully sponsored.

**Recognition**

- 2026: Best Alumni, AITU Awards.
- 2025: Best Student Researcher of the Year, AITU Awards.
- 2026: Winner, KMG Digital National Hackathon.
- 2026: Winner, Startup Battle, Digital Atyrau.
- 2025: Winner, Meta Llama Accelerator, Digital Bridge.
- 2025: Laureate, Golden Hephaestus Ministry Award for "Best Graduation Thesis of the Year" among Kazakhstan's national universities.
- 2025: Winner and Best Paper Award, IEEE Digital Generation 2025.
- 2025: Laureate, Digital Almaty Award for Best Innovative Solution in Big Data Analytics, awarded by the Ministry of Digital Development, Innovations and Aerospace Industry of the Republic of Kazakhstan.
- 2025: Winner, MultiDrone Championship, Digital Almaty.
- 2025: Winner, "Innovative Projects: A Look into the Future," International Student Project Competition (AGEU).
- 2025: Winner, "Management of Social Projects," International Student Project Competition (AGEU).
- 2025: Finalist, Jas Kenes Award, Council on Youth Policy under the President of Kazakhstan.
- 2025: Gold Award, Republican Student Research Competition, Ministry of Science and Higher Education of Kazakhstan.
- 2025: Winner, International Economics Olympiad, Toraighyrov University.
- 2025: Winner, Atyrau Youth Hackathon.
- 2025: Winner, International Scientific Conference, Narxoz University.
- 2025: Winner, QazDatathon, Bureau of National Statistics, Agency for Strategic Planning and Reforms.
- 2025: Winner, Google Developer Student Club Datathon, AITU.